

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

**INFR11157 NATURAL LANGUAGE UNDERSTANDING,
GENERATION, AND MACHINE TRANSLATION PG AND
INFR11225 UG**

Saturday 11th May 2024

13:00 to 15:00

INSTRUCTIONS TO CANDIDATES

1. Note that **ALL QUESTIONS ARE COMPULSORY.**
2. **DIFFERENT QUESTIONS MAY HAVE DIFFERENT NUMBERS OF TOTAL MARKS.** Take note of this in allocating time to questions.
3. This is a **NOTES NOT PERMITTED, CALCULATORS NOT PERMITTED** examination.

Notes and other written or printed material **MAY NOT BE CONSULTED** during the examination.

CALCULATORS MAY NOT BE USED IN THIS EXAMINATION.

Year 4 Courses

Convener: K.Etessami

External Examiners: J.Bowles, A.Pocklington, R.Mullins

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) Mary is interested in making her Transformer encoder block simpler. She decided to remove the multi-layered perceptron (MLP) layers and the layer normalisation parts. The only type of layer Mary wants to permit is the self-attention layer. She intends to stack such self-attention layers together (as a deep network) with a single output node that classifies text into its positive or negative sentiment.
 - i. Assuming the input embeddings are $x_1, \dots, x_n$, write explicit formulae (more than one formula) for a single layer of Mary's Transformer model that provides the output of that layer $y_1, \dots, y_n$. In your formulae, you may use *softmax* as a known function, and the key, query and value parameters, denoted by W_k , W_q and W_v . Briefly (one-two sentences) explain the reason to use the terms key, query and value in the context of Transformers. [4 marks]
 - ii. Beth and Mary had a discussion, in which Beth explained to Mary that it is likely not a good idea to simplify the Transformer as much and stack only self-attention layers. Write briefly one point of discussion Beth could have raised, possibly referring back to the formulae you stated in the answer to the previous question. [3 marks]
- (b) Ali wants to experiment with scaling laws for large language models, and partially replicate the experiment of the scaling laws of Chinchilla. As in class, let N be the number of parameters, D be the number of tokens used and $C(N, D)$ be the resulting compute needed. Ali tries to find a relationship between $L(N, D)$, the final training *loss*, and the parameters N and D . After having read the Chinchilla paper, he modelled $L(N, D)$ using the equation:

$$L(N, D) = \frac{a}{N^\alpha} + \frac{b}{D^\beta} + c. \quad (1)$$

- i. In his experiment, Ali found that the values for the parameters controlling $L(N, D)$ are: $\alpha = -0.2$ and $\beta = -0.6$, $a = 400$, $b = 400$, $c = 1.9$. Based on an analysis of the equation above, Ali realised he needs to rerun his experiment with different data or inspect his code, because some of these values are incorrect. Explain which of these values are likely incorrect, and provide a brief, one or two sentence explanation why. [3 marks]
- ii. What is the main measure of units that you expect Ali to use for $C(N, D)$? Briefly (and informally) explain what it calculates. [1 mark]

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(c) In class, we learned about an LSTM model that can be used to perform neural parsing (from the *Grammar as Foreign Language* paper). We learned that the tree output was linearised, and a sequence of nonterminals with opening brackets and closing brackets were being written by the decoder. Assume a simple LSTM model that outputs brackets in the form of (or), which unlike what we learned, does not specify a nonterminal with a closing bracket.

i. Rafael wants to find a simple solution to fix malformed trees that have more opening brackets than closing brackets. What is a simple post-processing step that could be used to fix the rare cases in which trees are malformed in that way?

[2 marks]

ii. Rafael decided that he wants to follow up on the original model we learned about, and this time, have different types of brackets being output together with their corresponding nonterminal (“type” the brackets). For example, we would have an opening bracket symbol such as (_{NP} or (_{VP}, and the corresponding closing brackets such as)_{NP} or)_{VP}. What new problem with brackets does this typing introduce that the solution in (i) will not solve, if we want to keep the opening/closing brackets matched with their corresponding nonterminals? Give a simple example or an illustration for a partially-linearised tree with “typed” brackets, that cannot be easily fixed.

[2 marks]

2. (a) We have a small parallel corpus of Spanish-English taken from a menu. We would like to learn a Spanish-English translation model over the data.

tortilla de patatas	- potato omelette
patatas bravas	- potato with spicy tomato sauce
pan con tomate	- bread with tomato

- i. Define a language model with a Markov assumption that models the joint probability of $P(\mathbf{y}, \mathbf{x})$, where Spanish sequence $x_1 \dots x_I$ is the input and $y_1 \dots y_K$ is the English output sequence. Do not use alignment. Describe one major disadvantage of your model. [2 marks]
- ii. You decide to use alignment to model translation and use expectation maximisation to learn the parameters of your n-gram translation model. Briefly write what you do in the expectation and maximisation steps. Explain in about one sentence why the parameter $p(\text{potato} \mid \text{patatas})$ increases after the first iteration. [2 marks]
- iii. You decide to implement a simple conditional RNN language model with no alignment as the translation model. You want the model to take the whole context of x and the previous target words y into account. You take this definition of a feedforward language model as your starting point. Re-write the whole model, making minimal changes to add one recurrence. The extra parameter matrix should be called $\mathbf{U}$ and the indexes are important.

$$\begin{aligned}
 P(y_{k+1} \mid x_{1:I}, x_I, y_k) &= \text{softmax}(\mathbf{V}\mathbf{h}_2 + \mathbf{b}_2) \\
 \mathbf{h}_2 &= \tanh(\mathbf{W}\mathbf{h}_1 + \mathbf{b}_1) \\
 \mathbf{h}_1 &= \mathbf{C}\mathbf{x}_{I-1}; \mathbf{C}\mathbf{x}_I; \mathbf{C}\mathbf{y}_k \\
 \mathbf{y}_k &= \text{onehot}(y_k)
 \end{aligned}$$

- iv. What linguistic/contextual feature could be added to the RNN model improve performance? [3 marks]
- v. We want to learn a BPE merge model over just the Spanish part of our menu dataset. Please write down the first merge. Only consider merges that occur within words, and do not consider word boundaries in the calculation. [1 mark]
- vi. Briefly describe what is the advantage of using BPE-dropout? [1 mark]

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- (b) You want to develop a machine translation system for Wolof→English. Wolof is spoken by over 5 million people in Senegal, the Gambia, and Mauritania, but there is very little data for training the model, only a few thousand sentences of Wolof-English translation pairs. You do however have access to a Wolof corpus taken from Wikipedia with about 100,000 sentences. None of these questions should require more than a few sentences.
- i. You train a Wolof→English machine translation encoder-decoder model on the parallel data but performance is terrible. How can you use the monolingual data to improve performance? Please write about two sentences, each containing one idea that will address this question. How can you use data from other parallel datasets (which will likely contain English but will not contain Wolof) to improve performance? Explain one idea that will address this question. *[3 marks]*
 - ii. You decide to explore the Wolof→English translation performance of a large language model (LLM), but no LLMs have been trained with Wolof data. How can you fine-tune an LLM to be able to translate from Wolof→English? Please write three high-level ideas to address this problem. *[3 marks]*
 - iii. Describe the input and output of the COMET score. *[1 mark]*
 - iv. Why could the COMET score be unreliable for testing an English→Wolof translation model? Would there be any difference in COMET reliability when evaluating a Wolof→English system? *[2 marks]*
 - v. If an immigration service were to rely on this automatic translation model to translate asylum claims, describe how this could cause harm, and what type of harm could it cause. *[1 mark]*

3. (a) You are developing a question-answering (QA) system. Your task is to design an efficient way to provide context to the model that improves its ability to answer questions about a specific domain. More specifically, you are focusing on *trivia question answering*: given a trivia question, the model needs to produce the corresponding answer.

Example 1 (Input/Output Pairs) Refer to the following three examples for a possible input (a question) and the desired output (the corresponding answer) for your model.

Input: In what year did World War II end?
Desired model output: 1945.

Input: Who painted the Mona Lisa?
Desired model output: Leonardo da Vinci.

Input: How many elements are there in the periodic table?
Desired model output: 118.

The model needs to answer each question without any supporting document or paragraph — we are in a *open-domain question answering* setting.

- i. You are implementing the QA system using in-context learning (ICL) with an underlying large language model (LLM), for example, LLAMA2 70B. Describe how you would structure the input to the model to maximize the effectiveness of in-context learning for this domain-specific QA system: write about two sentences about how you would prompt the model for implementing a QA system using ICL, describing the format of the input and its characteristics, and then provide a brief example. Describe how you would select and format the context provided to the model; you can illustrate your approach using the examples above. Finally, discuss a potential challenge of using in-context learning for domain-specific QA regarding context length. [4 marks]
- ii. In addition to in-context learning, you also want to try implementing the QA system by performing a (full) fine-tuning of the parameters of the LLM. How do you define the input and output data for the fine-tuning? What kind of training objective would you use? How would you decide which system to use in production — your system trained via in-context learning or the one trained via (full) fine-tuning? [4 marks]

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(b) Parameter-efficient fine-tuning methods, such as *adapter modules*, have been proposed to customize pre-trained LLMs for specific tasks without the need to fine-tune the entire model. Assume you have decided to use LoRA to fine-tune a pre-trained model for your domain-specific QA system.

- i. Explain the concept of adapter modules and how they can be integrated into a pre-trained model for fine-tuning a pre-trained LLM on the QA task; write two or three sentences. Write the equations of the LoRA adapter and how it can be used for updating the key, query, and value parameters, denoted by W_k , W_q , and W_v , and describe them in one or two sentences.

[4 marks]

- ii. Compare the computational costs of inference and training of using adapter modules for fine-tuning on the QA task, as opposed to full model fine-tuning; answer in two or three sentences. Then, discuss how in-context learning can contribute to reducing the computational resources required for developing and deploying NLP models — answer briefly in one or two sentences.

[3 marks]